Math 54 — Week 7

Coordinate systems; change of basis; eigenvectors and eigenvalues; the characteristic equation (Lay 4.4, 4.6, 5.1–5.2)

§1 Facts used from Weeks 1–6

(W1) A basis of V is an independent spanning set; every basis of V has $\dim V$ vectors. With $\dim V = n$, n independent vectors are a basis.

(W2) Invertible matrix theorem: for square A, invertible $\iff$ $A\mathbf{x} = \mathbf{0}$ only trivially $\iff$ $\det A \neq 0$ $\iff$ columns form a basis of $\mathbb{R}^n$.

(W3) $\det(AB) = \det A \det B$; det of a triangular matrix is the product of its diagonal entries.

(W4) $\operatorname{Nul} A$ is a subspace; a basis comes from the parametric vector form of $A\mathbf{x} = \mathbf{0}$. A one-to-one linear map sends only $\mathbf{0}$ to $\mathbf{0}$.

§2 This week

§2.1 Coordinate systems (Lay 4.4)

The question. A basis lets us describe each vector of V by a list of numbers. Why is that list unique, and how do we compute it?

Theorem (Unique representation)
If $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ is a basis of V, each $\mathbf{x}$ in V can be written as $\mathbf{x} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n$ in exactly one way.

Proof. Spanning gives at least one way. If also $\mathbf{x} = d_1\mathbf{b}_1 + \cdots + d_n\mathbf{b}_n$, subtract: $\mathbf{0} = (c_1 - d_1)\mathbf{b}_1 + \cdots + (c_n - d_n)\mathbf{b}_n$. Independence gives $c_i = d_i$ for every i. □

Definition. These weights are the **coordinates of $\mathbf{x}$ relative to $\mathcal{B}$**: $[\mathbf{x}]_{\mathcal{B}} = (c_1, \ldots, c_n) \in \mathbb{R}^n$. The map $\mathbf{x} \mapsto [\mathbf{x}]_{\mathcal{B}}$ is the **coordinate mapping**.

Coordinates in $\mathbb{R}^n$. Let $P_{\mathcal{B}} = [\,\mathbf{b}_1 \cdots \mathbf{b}_n\,]$, the **change-of-coordinates matrix** from $\mathcal{B}$ to the standard basis. Then

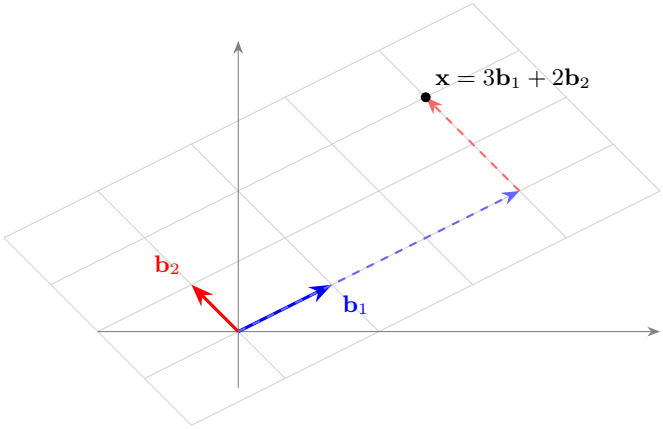
$$\mathbf{x} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n = P_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}, \qquad \text{so} \qquad [\mathbf{x}]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}\mathbf{x}.$$

($P_{\mathcal{B}}$ is invertible because its columns are a basis of $\mathbb{R}^n$.) In practice, find $[\mathbf{x}]_{\mathcal{B}}$ by row reducing $[\,\mathbf{b}_1 \cdots \mathbf{b}_n \;\; \mathbf{x}\,]$.

Worked example. $\mathbf{b}_1 = (2, 1)$, $\mathbf{b}_2 = (-1, 1)$, $\mathbf{x} = (4, 5)$. Solve $c_1\mathbf{b}_1 + c_2\mathbf{b}_2 = \mathbf{x}$:

$$\left[\begin{array}{cc|c} 2 & -1 & 4 \\ 1 & 1 & 5 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 2 \end{array}\right], \qquad [\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Check: $3(2, 1) + 2(-1, 1) = (4, 5)$.



The grid of the basis $\mathcal{B}$: go 3 steps along $\mathbf{b}_1$, then 2 steps along $\mathbf{b}_2$. In standard coordinates the same point is $(4, 5)$.

Theorem (The coordinate mapping is an isomorphism)

If $\mathcal{B}$ is a basis of V with n vectors, $\mathbf{x} \mapsto [\mathbf{x}]_{\mathcal{B}}$ is a one-to-one linear transformation from V onto $\mathbb{R}^n$.

Proof. 1. *Linear.* If $\mathbf{u} = \sum c_i \mathbf{b}_i$ and $\mathbf{w} = \sum d_i \mathbf{b}_i$, then $\mathbf{u} + \mathbf{w} = \sum (c_i + d_i)\mathbf{b}_i$ and $r\mathbf{u} = \sum (rc_i)\mathbf{b}_i$. By uniqueness these are the coordinates: $[\mathbf{u} + \mathbf{w}]_{\mathcal{B}} = [\mathbf{u}]_{\mathcal{B}} + [\mathbf{w}]_{\mathcal{B}}$ and $[r\mathbf{u}]_{\mathcal{B}} = r[\mathbf{u}]_{\mathcal{B}}$.

2. *One-to-one.* If $[\mathbf{x}]_{\mathcal{B}} = \mathbf{0}$, then $\mathbf{x} = 0\mathbf{b}_1 + \cdots + 0\mathbf{b}_n = \mathbf{0}$.
3. *Onto.* Any $(c_1, \ldots, c_n)$ is the coordinate vector of $c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n$.

□

A one-to-one linear map from V onto W is an **isomorphism**. The theorem says every n-dimensional space behaves exactly like $\mathbb{R}^n$: questions about polynomials or matrices become questions about columns of numbers.

Using it: independence of polynomials. Are $p_1 = 1 + 2t^2$, $p_2 = 4 + t + 5t^2$, $p_3 = 3 + 2t$ independent in $\mathbb{P}_2$? With the standard basis $\{1, t, t^2\}$ their coordinate vectors are $(1, 0, 2)$, $(4, 1, 5)$, $(3, 2, 0)$. A linear relation among the polynomials is the same as one among their coordinate vectors (the coordinate map is linear and one-to-one).

$$\begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 2 \\ 2 & 5 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Column 3 is -5 (column 1) $+2$ (column 2), so $p_3 = -5p_1 + 2p_2$: dependent. Check: $-5 - 10t^2 + 8 + 2t + 10t^2 = 3 + 2t$.

§2.2 Change of basis (Lay 4.6)

The question. With two bases $\mathcal{B}$ and $\mathcal{C}$ of the same space, how do we convert $\mathcal{B}$-coordinates into $\mathcal{C}$-coordinates directly?

Theorem (Change-of-coordinates matrix)

Let $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ and $\mathcal{C}$ be bases of V. Then $[\mathbf{x}]_{\mathcal{C}} = P_{\mathcal{C} \leftarrow \mathcal{B}} [\mathbf{x}]_{\mathcal{B}}$ for every $\mathbf{x}$, where

$$P_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{bmatrix} [\mathbf{b}_1]_{\mathcal{C}} & [\mathbf{b}_2]_{\mathcal{C}} & \cdots & [\mathbf{b}_n]_{\mathcal{C}} \end{bmatrix}.$$

It is invertible, and $(P_{\mathcal{C} \leftarrow \mathcal{B}})^{-1} = P_{\mathcal{B} \leftarrow \mathcal{C}}$.

The idea

Write $\mathbf{x} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n$, so $[\mathbf{x}]_{\mathcal{B}} = (c_1, \ldots, c_n)$. Taking $\mathcal{C}$-coordinates is linear, so $[\mathbf{x}]_{\mathcal{C}} = c_1[\mathbf{b}_1]_{\mathcal{C}} + \cdots + c_n[\mathbf{b}_n]_{\mathcal{C}}$. That is a matrix times $(c_1, \ldots, c_n)$, and the matrix has the columns $[\mathbf{b}_j]_{\mathcal{C}}$. So you only need to express the *old* basis vectors in the *new* basis.

Proof. The computation in the idea box is the proof. The columns of $P_{\mathcal{C}\leftarrow\mathcal{B}}$ are the images of the independent vectors $\mathbf{b}_j$ under the one-to-one coordinate map, so they are independent and $P_{\mathcal{C}\leftarrow\mathcal{B}}$ is invertible. Multiplying $[\mathbf{x}]_{\mathcal{C}} = P_{\mathcal{C}\leftarrow\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}$ by its inverse gives the conversion in the other direction. □

Computing $P_{\mathcal{C}\leftarrow\mathcal{B}}$ in $\mathbb{R}^n$. Each column $[\mathbf{b}_j]_{\mathcal{C}}$ solves $[\mathbf{c}_1 \cdots \mathbf{c}_n]\mathbf{y} = \mathbf{b}_j$. Solve all of them at once:

$$[\,\mathbf{c}_1 \;\cdots\; \mathbf{c}_n \;\mid\; \mathbf{b}_1 \;\cdots\; \mathbf{b}_n\,] \longrightarrow [\,I \;\mid\; P_{\mathcal{C}\leftarrow\mathcal{B}}\,].$$

Worked example. $\mathcal{B} = \{(1,1),(1,2)\}$, $\mathcal{C} = \{(1,0),(2,1)\}$.

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{array}\right] \xrightarrow{R_1-2R_2} \left[\begin{array}{cc|cc} 1 & 0 & -1 & -3 \\ 0 & 1 & 1 & 2 \end{array}\right], \qquad P_{\mathcal{C}\leftarrow\mathcal{B}} = \begin{bmatrix} -1 & -3 \\ 1 & 2 \end{bmatrix}.$$

Check the first column: $-1(1,0) + 1(2,1) = (1,1) = \mathbf{b}_1$. The inverse is $P_{\mathcal{B}\leftarrow\mathcal{C}} = \begin{bmatrix} 2 & 3 \\ -1 & -1 \end{bmatrix}$.

- With $\mathcal{C}$ the standard basis $\mathcal{E}$, $P_{\mathcal{E}\leftarrow\mathcal{B}} = P_{\mathcal{B}}$, the matrix from 4.4.
- Also $P_{\mathcal{C}\leftarrow\mathcal{B}} = P_{\mathcal{C}}^{-1}P_{\mathcal{B}}$: go from $\mathcal{B}$ to standard coordinates, then from standard to $\mathcal{C}$.

§2.3 Eigenvectors and eigenvalues (Lay 5.1)

The question. A matrix moves most vectors off their line through the origin. Which vectors does it only stretch, and why are they the right tool for understanding $A^k\mathbf{x}$?

Definition. An **eigenvector** of an $n \times n$ matrix A is a *nonzero* vector $\mathbf{x}$ with $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ, the **eigenvalue**. For an eigenvalue λ, the set of all solutions of $(A - \lambda I)\mathbf{x} = \mathbf{0}$ is the **eigenspace** of A for λ; it is $\mathrm{Nul}(A - \lambda I)$, a subspace consisting of $\mathbf{0}$ and all eigenvectors for λ.

- **λ is an eigenvalue $\iff$ $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution.** Because $A\mathbf{x} = \lambda\mathbf{x}$ is the same equation as $A\mathbf{x} - \lambda I\mathbf{x} = \mathbf{0}$.
- **Checking a vector**: compute $A\mathbf{u}$ and see if it is a multiple of $\mathbf{u}$. With $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$: $A(6,-5) = (-24,20) = -4(6,-5)$, so $(6,-5)$ is an eigenvector for -4.
- **Checking a number**: row reduce $A - \lambda I$ and look for a free variable.

Worked example: an eigenspace. Is $\lambda = 2$ an eigenvalue of the matrix below, and what is its eigenspace?

$$A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}, \qquad A - 2I = \begin{bmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -1 & 6 \end{bmatrix} \to \begin{bmatrix} 2 & -1 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Two free variables, so 2 is an eigenvalue. $x_1 = \frac{1}{2}x_2 - 3x_3$:

$$\mathbf{x} = x_2\begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix} + x_3\begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}, \qquad \text{basis of the eigenspace: } \{(1,2,0),\ (-3,0,1)\}.$$

Geometrically: A multiplies every vector in this plane by 2.

Theorem (Eigenvalues of a triangular matrix)
The eigenvalues of a triangular matrix are the entries on its main diagonal.

Proof. If A is triangular, so is $A - \lambda I$, with diagonal entries $a_{ii} - \lambda$. $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution $\iff$ $A - \lambda I$ is not invertible $\iff$ its determinant $\prod(a_{ii} - \lambda)$ is 0 $\iff$ $\lambda = a_{ii}$ for some i. □

- **0 is an eigenvalue of A $\iff$ A is not invertible** ($A\mathbf{x} = 0\mathbf{x}$ has a nontrivial solution).

Theorem (Eigenvectors for distinct eigenvalues are independent)

If $\mathbf{v}_1, \dots, \mathbf{v}_r$ are eigenvectors for distinct eigenvalues $\lambda_1, \dots, \lambda_r$, then $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is linearly independent.

The idea

Suppose not, and take the first $\mathbf{v}_{p+1}$ that is a combination of the earlier ones: $\mathbf{v}_{p+1} = c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p$, with $\mathbf{v}_1, \dots, \mathbf{v}_p$ independent. Applying A and multiplying by λ_{p+1} do the same thing to $\mathbf{v}_{p+1}$ but different things to the other vectors. Subtracting the two results removes $\mathbf{v}_{p+1}$ and leaves a relation among $\mathbf{v}_1, \dots, \mathbf{v}_p$ whose weights are $c_i(\lambda_i - \lambda_{p+1})$; independence forces every $c_i = 0$.

Proof. 1. If the set is dependent, there is a least index p such that $\mathbf{v}_{p+1}$ is a combination of $\mathbf{v}_1, \dots, \mathbf{v}_p$; by the choice of p, $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is independent. Write $\mathbf{v}_{p+1} = c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p$.
2. Multiply by A: $\lambda_{p+1}\mathbf{v}_{p+1} = c_1\lambda_1\mathbf{v}_1 + \cdots + c_p\lambda_p\mathbf{v}_p$.
3. Multiply step 1 by λ_{p+1}: $\lambda_{p+1}\mathbf{v}_{p+1} = c_1\lambda_{p+1}\mathbf{v}_1 + \cdots + c_p\lambda_{p+1}\mathbf{v}_p$.
4. Subtract: $\mathbf{0} = c_1(\lambda_1 - \lambda_{p+1})\mathbf{v}_1 + \cdots + c_p(\lambda_p - \lambda_{p+1})\mathbf{v}_p$. By independence each $c_i(\lambda_i - \lambda_{p+1}) = 0$, and the eigenvalues are distinct, so each $c_i = 0$.
5. Then $\mathbf{v}_{p+1} = \mathbf{0}$, which an eigenvector is not. Contradiction.

□

Eigenvectors and difference equations. For $\mathbf{x}_{k+1} = A\mathbf{x}_k$: if $\mathbf{x}_0$ is an eigenvector with eigenvalue λ, then $\mathbf{x}_k = \lambda^k\mathbf{x}_0$. The direction never changes; only the length is multiplied by λ each step. Next week (diagonalization) every starting vector is split into such pieces.

§2.4 The characteristic equation (Lay 5.2)

The question. Checking one candidate λ at a time is slow. How do we find all eigenvalues at once?

Theorem (The characteristic equation)

λ is an eigenvalue of the $n \times n$ matrix A if and only if $\det(A - \lambda I) = 0$.

Proof. λ is an eigenvalue $\iff$ $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution $\iff$ $A - \lambda I$ is not invertible $\iff$ $\det(A - \lambda I) = 0$ (W2). □

$\det(A - \lambda I)$ is a polynomial of degree n in λ, the **characteristic polynomial**. The **algebraic multiplicity** of an eigenvalue is its multiplicity as a root.

Worked example (2×2). $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$:

$$\det \begin{bmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{bmatrix} = (3-\lambda)^2 - 1 = \lambda^2 - 6\lambda + 8 = (\lambda - 2)(\lambda - 4).$$

Eigenvalues 2 and 4. For $\lambda = 2$: $A - 2I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, eigenvector $(-1, 1)$. For $\lambda = 4$: $A - 4I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$, eigenvector $(1, 1)$.

Worked example (3×3). Expand $\det(A - \lambda I)$ down column 1, which has one nonzero entry:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 0 \\ 0 & 1 & 2 \end{bmatrix}, \qquad \det(A - \lambda I) = (1-\lambda)\det \begin{bmatrix} 3-\lambda & 0 \\ 1 & 2-\lambda \end{bmatrix} = (1-\lambda)(3-\lambda)(2-\lambda).$$

Eigenvalues $1, 2, 3$, each of multiplicity one.

For a 2×2 matrix, $\det(A - \lambda I) = \lambda^2 - (\operatorname{tr} A)\lambda + \det A$, where $\operatorname{tr} A = a + d$ is the sum of the diagonal entries. So the two eigenvalues add up to $a + d$ and multiply to $ad - bc$.

Definition. A and B are **similar** if $B = P^{-1}AP$ for some invertible P.

Theorem (Similar matrices have the same characteristic polynomial)
If $B = P^{-1}AP$, then A and B have the same characteristic polynomial, hence the same eigenvalues with the same multiplicities.

Proof. $B - \lambda I = P^{-1}AP - \lambda P^{-1}P = P^{-1}(A - \lambda I)P$. So $\det(B - \lambda I) = \det(P^{-1}) \det(A - \lambda I) \det P = \det(A - \lambda I)$, because $\det(P^{-1}) \det P = 1$. □

- The converse is false: $\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ and $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ have the same characteristic polynomial but are not similar ($P^{-1}(2I)P = 2I$ for every P).
- Similarity is not row equivalence. Row operations usually change eigenvalues.
- **Added to the invertible matrix theorem**: A is invertible $\iff$ 0 is not an eigenvalue $\iff$ $\det A \neq 0$.

Shortcuts

- **Coordinates in $\mathbb{R}^n$**: row reduce $[\,\mathbf{b}_1 \cdots \mathbf{b}_n \;\mathbf{x}\,]$. In $\mathbb{P}_n$ or matrix spaces: use standard coordinates and work with columns.
- **Change of basis**: $[\,\mathcal{C} \mid \mathcal{B}\,] \to [\,I \mid P_{\mathcal{C}\leftarrow\mathcal{B}}\,]$. Columns of $P_{\mathcal{C}\leftarrow\mathcal{B}}$ are old basis vectors in new coordinates.
- **Is u an eigenvector?** One multiplication. **Is λ an eigenvalue?** One row reduction of $A - \lambda I$.
- **Triangular**: eigenvalues are the diagonal entries; no determinant needed.
- 2×2: $\lambda^2 - (a+d)\lambda + (ad - bc)$; eigenvalues sum to the trace and multiply to the determinant. An eigenvector for λ: $(b,\ \lambda - a)$ if $b \neq 0$.
- 3×3 **characteristic polynomial**: expand $\det(A - \lambda I)$ along a row or column with zeros, and *keep factors* such as $(1 - \lambda)$ instead of multiplying out.
- **Powers and inverses**: if $A\mathbf{x} = \lambda\mathbf{x}$ then $A^k\mathbf{x} = \lambda^k\mathbf{x}$, and if A is invertible $A^{-1}\mathbf{x} = \lambda^{-1}\mathbf{x}$.

Traps

- An eigenvector must be nonzero; an eigenvalue may be 0.
- The eigenspace is $\mathrm{Nul}(A - \lambda I)$; row reduce $A - \lambda I$, not A.
- Row reducing A does not preserve its eigenvalues. Never read eigenvalues off an echelon form of A.
- The order matters in $P_{\mathcal{C}\leftarrow\mathcal{B}}$: the columns are the $\mathcal{B}$ vectors written in $\mathcal{C}$.
- Same characteristic polynomial does not mean similar.
- $\det(A - \lambda I)$, not $\det A - \lambda$.

§3 Homework this week

HW6, Section 4.4: 1, 3, 7, 13, 15, 17, 19, 25, 36. Section 4.6: 1, 5, 6, 9, 12, 13, 15. HW7, Section 5.1: 3, 5, 7, 9, 13, 23, 24, 27, 28, 29, 30, 34. Section 5.2: 5, 7, 11, 13, 15, 18, 19, 24, 25.

§4 Problem set

P1. Find $[\,\mathbf{x}\,]_{\mathcal{B}}$ for $\mathbf{x} = (4, 9)$ and $\mathcal{B} = \{(1, 3), (2, 5)\}$.

P2. (a) $\mathcal{B} = \{1 + t,\ t + t^2,\ 1 + t^2\}$ is a basis of $\mathbb{P}_2$. Find $[\,p\,]_{\mathcal{B}}$ for $p = 3 + 2t - t^2$. (b) Use coordinate vectors to decide whether $1 - t$, $t - t^2$, $1 - t^2$ are linearly independent.

P3. $\mathcal{B} = \{(1, 2), (3, 4)\}$ and $\mathcal{C} = \{(1, 1), (0, 1)\}$. Find $P_{\mathcal{C}\leftarrow\mathcal{B}}$ and $P_{\mathcal{B}\leftarrow\mathcal{C}}$. If $[\,\mathbf{x}\,]_{\mathcal{B}} = (2, -1)$, find $[\,\mathbf{x}\,]_{\mathcal{C}}$.

P4. Let $A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$. Is $\mathbf{v} = (-1, 1, 1)$ an eigenvector of A? If so, for which eigenvalue? Is 4 an eigenvalue of A?

P5. For the same A, find the characteristic polynomial, all eigenvalues, and a basis of the eigenspace for $\lambda = 2$.

P6. Find the characteristic polynomials and eigenvalues of $\begin{bmatrix} 2 & 7 \\ 7 & 2 \end{bmatrix}$ and $\begin{bmatrix} 5 & -3 \\ -4 & 3 \end{bmatrix}$.

P7. List the eigenvalues, with multiplicities, of

$$\begin{bmatrix} 5 & -2 & 6 & -1 \\ 0 & 3 & -8 & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

P8. Suppose $A\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$. (a) Show that $\mathbf{x}$ is an eigenvector of A^2 and of $A + 3I$; for which eigenvalues? (b) If A is invertible, show $\lambda \neq 0$ and $A^{-1}\mathbf{x} = \lambda^{-1}\mathbf{x}$.